Self-Supervised Learning for Electricity-Theft Detection (SGCC)

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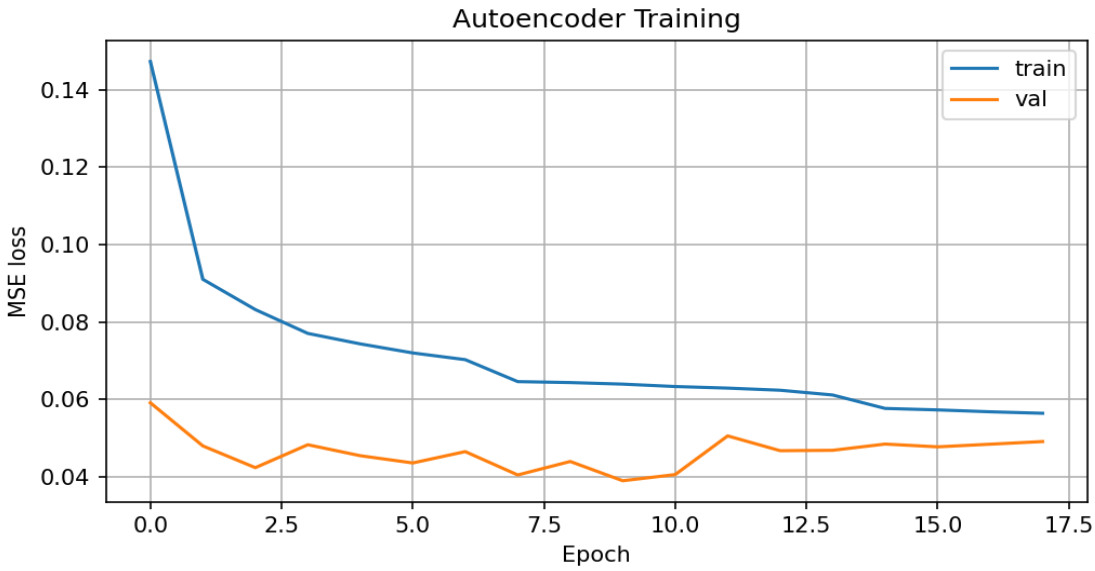
Objective: Detect electricity theft patterns using a self-supervised denoising autoencoder trained on sliding windows of daily smart-meter data.

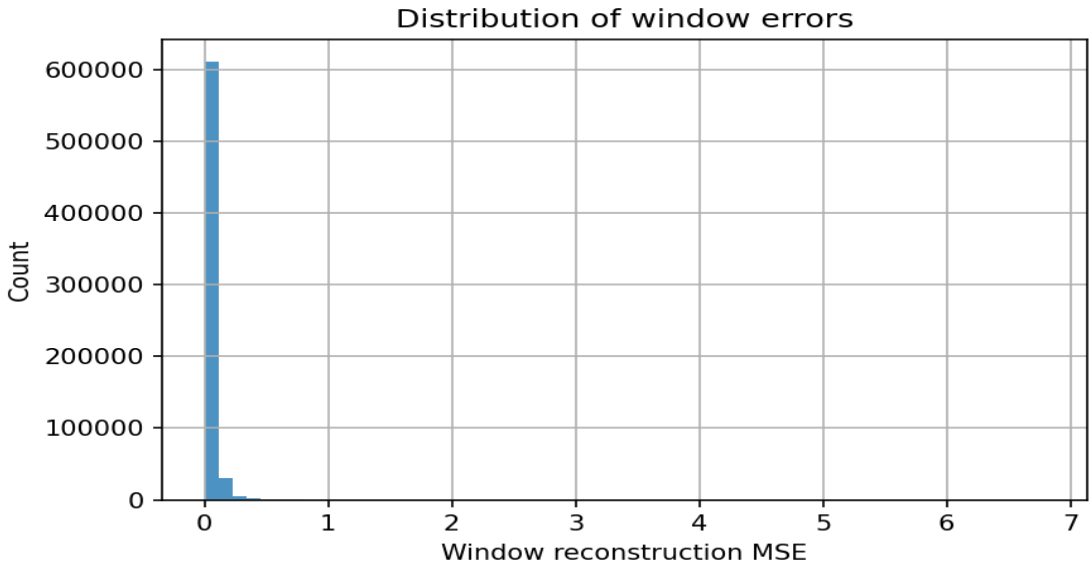
Pipeline (summary): daily aggregation → per-customer z-score → windowing (L=60, stride=15) → SSL pretext (reconstruction) → window MSE → 95th-pct customer score.

Customers scored	10,000
Threshold method	Quantile = 0.975 (thr = 0.313880)
Suspects ≥ threshold	250
Artifacts dir	/content/figs

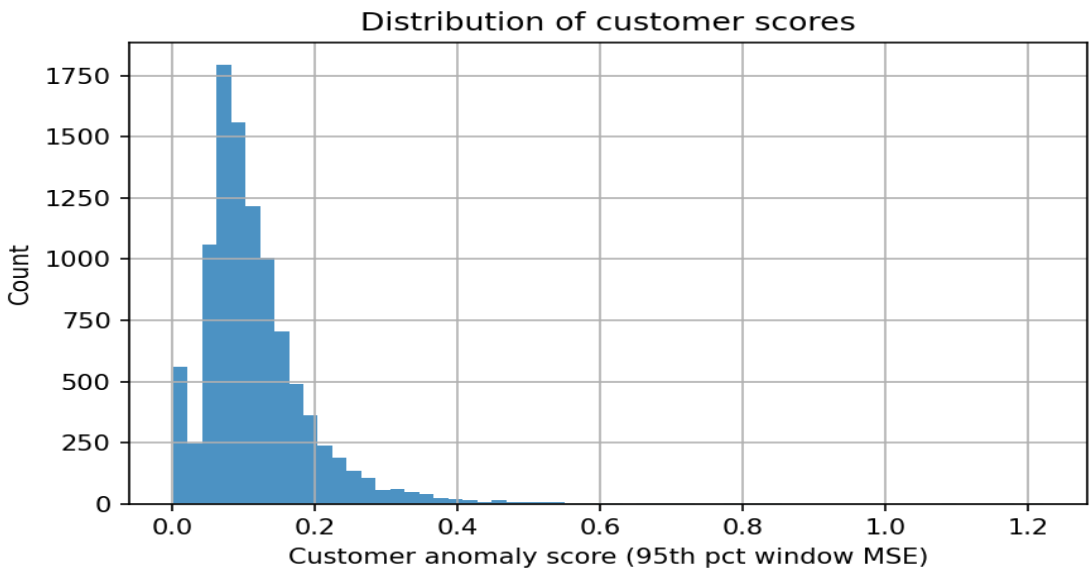
Training Curve



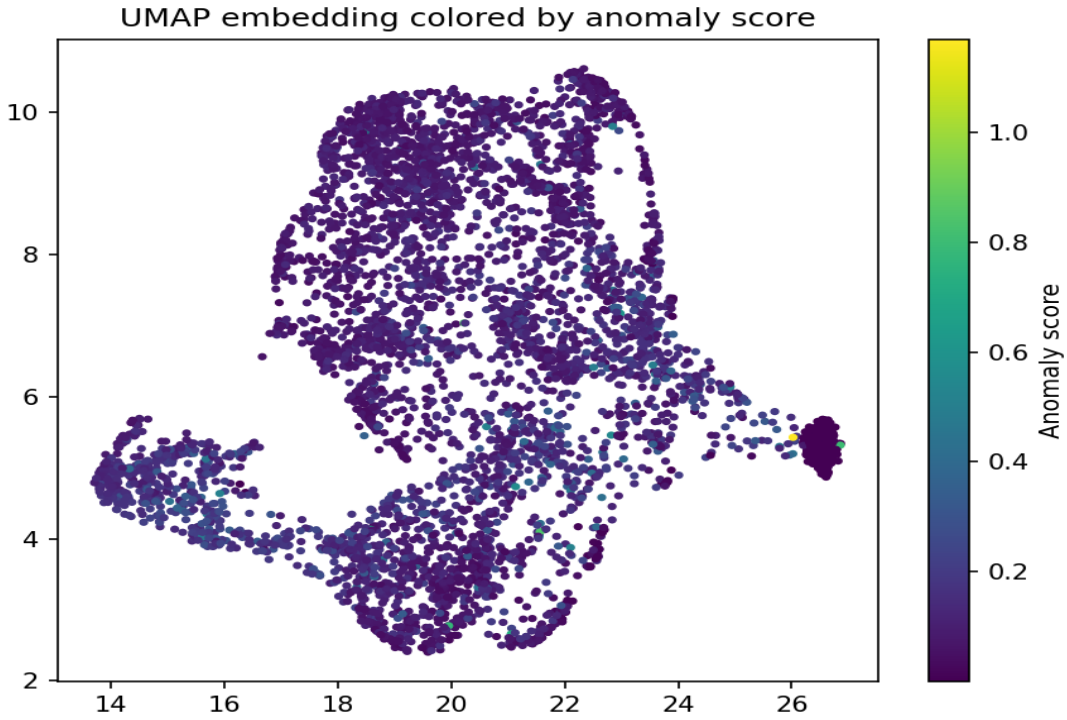
Distribution of Window Reconstruction Errors



Distribution of Customer Anomaly Scores



Customer Embedding (PCA→UMAP) colored by anomaly score


UMAP embedding colored by anomaly score
10
8
6
4
2
14
16
18
20
22
24
26
1.0
0.8
0.6
0.4
0.2
Anomaly score

Top Suspected Customers

Rank	Customer ID	Anomaly Score
1	3CA604D6F189397154C106D2397424D3	1.220475
2	DF84CF0138D8ADE49D843DFCEDB431EF	1.169536
3	93B720F500AC52EC5F8AF30249BC7021	1.163515
4	4AD1E63C55BC42B5ADD7585940BD4A44	1.080573
5	FF2480F5CB1925C77859FBA1538B7256	0.851530
6	6BBDA619B79E28083E6A386CF432B89B	0.843028
7	6F739D77D52009669991F6C44E96005D	0.836662
8	3323053BE89AC082F6C93D11648902DF	0.822771
9	6E15271FB5136A23A9F7A333D2821B73	0.802135
10	8F2845B313DB6D103C77ED94C2C82AF7	0.797073
11	A70828D956AAFFAADC2457BAD81F8DCF	0.780691
12	0E4BE6D75AA4AAA1E5EDFD40E30FC59C	0.727698
13	AA5A5AC4E89ED83D62B67F3DA23F6111	0.718590
14	7EC081AA42AAA6D73652F302F8750621	0.697151
15	83DA511EBD8647171EA87B6CCC37F6F9	0.691427
16	0F26CB84297DAA19D5314F1273D5A053	0.681151
17	5866CCB423FD49E0BAC6473AD6377058	0.672591
18	B94B652C3C0488463FB1154849BA7D33	0.653212
19	2E7E8136AE4182BD88EFD17EB76EBCA6	0.652465
20	42F8B42DCC7B2201D3B5016DC0DE37F7	0.611255
21	33F2073BB5348AFF8D082332ACCAB61B	0.603929
22	8FFB2E9B2F4C8F0000AB995AC23EC307	0.578972
23	545904B04AC6308131B3769E8E5C8C70	0.570562
24	FE7828AAADF3771C243B5F9ADF106E33	0.568688
25	1985D41B42BBF2CB50CA8EA749061CD8	0.562324
26	7BCF3DC5BE4DEDBBBA1B7728E91E45D8	0.557231
27	6E48067A5C07A99338273F5CBC59E443	0.544460
28	3B591D40E0CF49024AA5EFD90E68779F	0.544033
29	DB3B61D7C352141850F38BFD7FCF7709	0.543730
30	821ADFD76B6FCD6886519192D7E2A4E5	0.541040
31	F1120F707B045A3AC9DE2ABA2D5BB1A6	0.533237
32	EF943DC98D4533ADF8275CA244468078	0.530702
33	24F8FC1A4EACC4B75270C5BE6E3B087A	0.525221
34	A09E2B59AC260E1A279B344787D19428	0.524463
35	3AA2D7C17CF6A05C78A3B6DDBA69F977	0.514337
36	2AF257FCFA84BE459E7B01F55EFE8CEB	0.513050
37	9E3B0E1EAD96EC81BE330B412BB7A879	0.511003
38	3C69EBE30ECADB0BDED9225D094A8DF6	0.502088

39	C8BC2D024C8DD2B25A631B895B80FD19	0.500754
40	DE7831F40E6BF7324828778415B97CD2	0.500495
41	AA468673AB9C4EC86438A0405E3EB630	0.496597
42	8F3F21B5495771110E06DBCDF02F7971	0.492645
43	033A4E4B5817F014486061B4B2F065EC	0.492513
44	2206F5C51B2B9A7E3ACFCC4EA72F67E2	0.489754
45	0E92F2C7E44E214BC7D7F885F8F86174	0.486971
46	82B667CAA679C4B0AA132578FEB9B603	0.486569
47	FABD1514F34938D6F92A9EF58D47C409	0.486282
48	022C4693991377D851CAA1509451219C	0.483514
49	477B410B27B2098F8123246DA01BDAD9	0.478171
50	D7762CE3F8A08E8D0C630AD90AA5C1E9	0.474165

For any listed customer, you can plot its window-error time series using the provided sanity-check cell (see notebook) to visually verify localized anomalies.